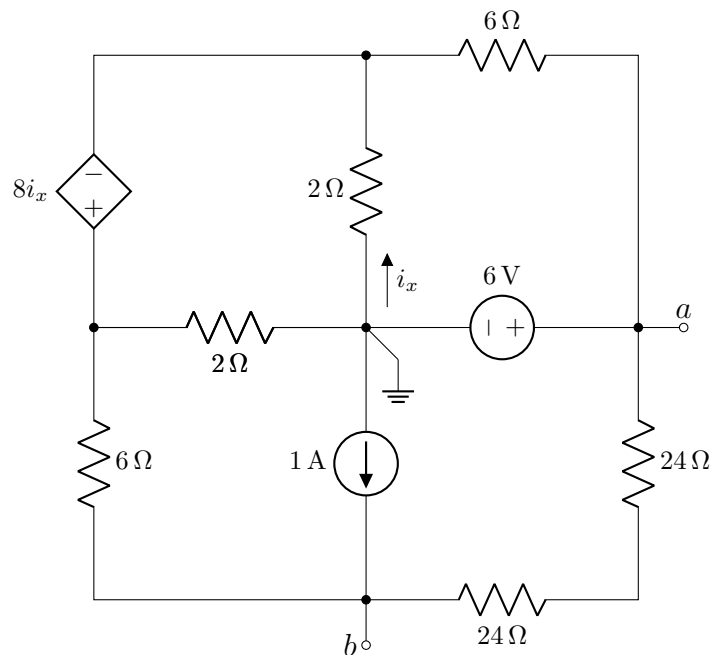


- ✓ No washroom breaks. Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to invigilator.
- ✓ All **3 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Symbols have their usual meanings.

■ Question 1 of 3 [CO2 CO3 CO4] [20 marks]

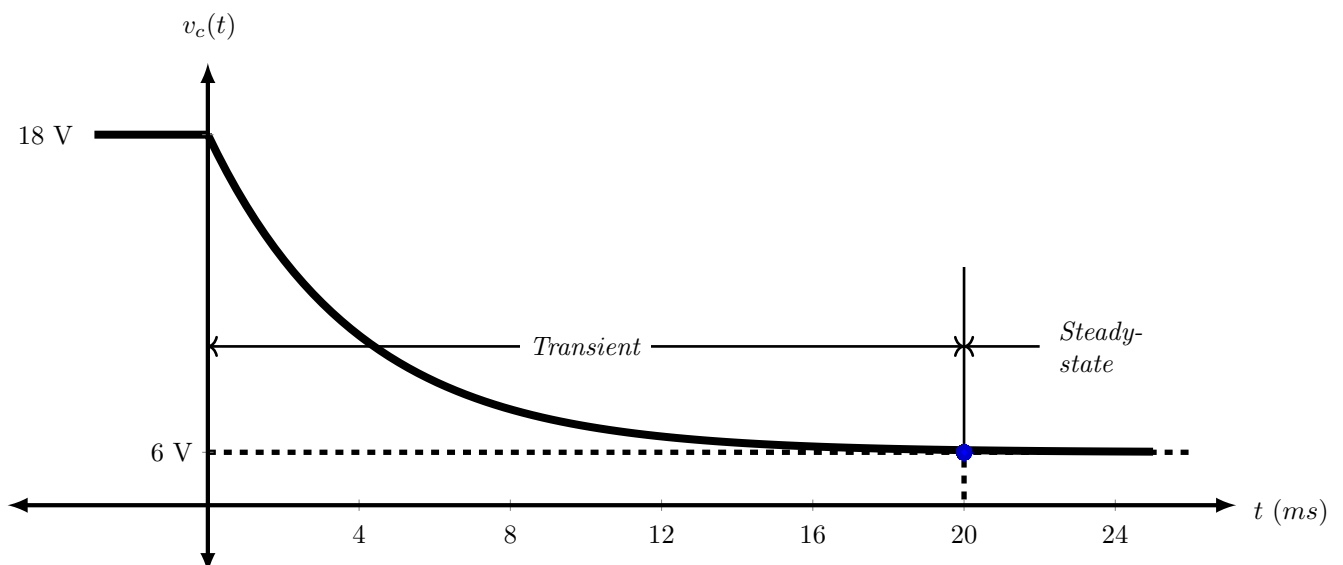
Consider the following circuit with open terminals *a* and *b*. Currently, no load is connected to the terminals.



- (a) [12 marks] Determine the value of R_L that will draw the **Maximum Power** from the circuit.
- (b) [8 marks] Determine the value of the **Maximum Power**.

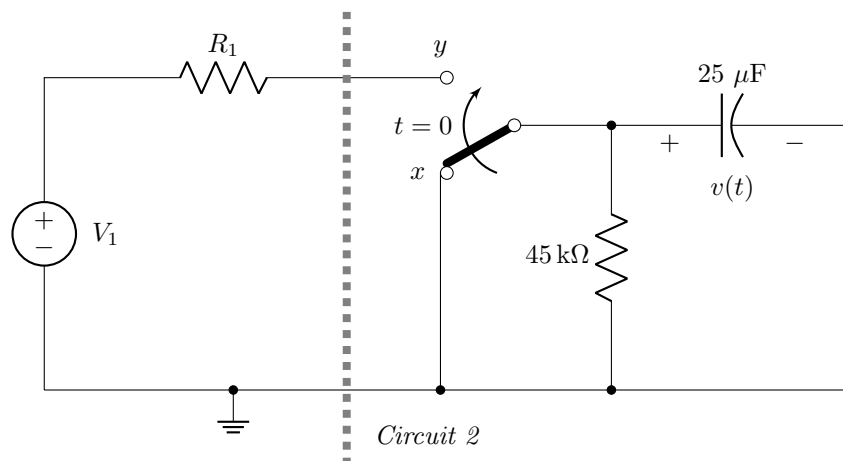
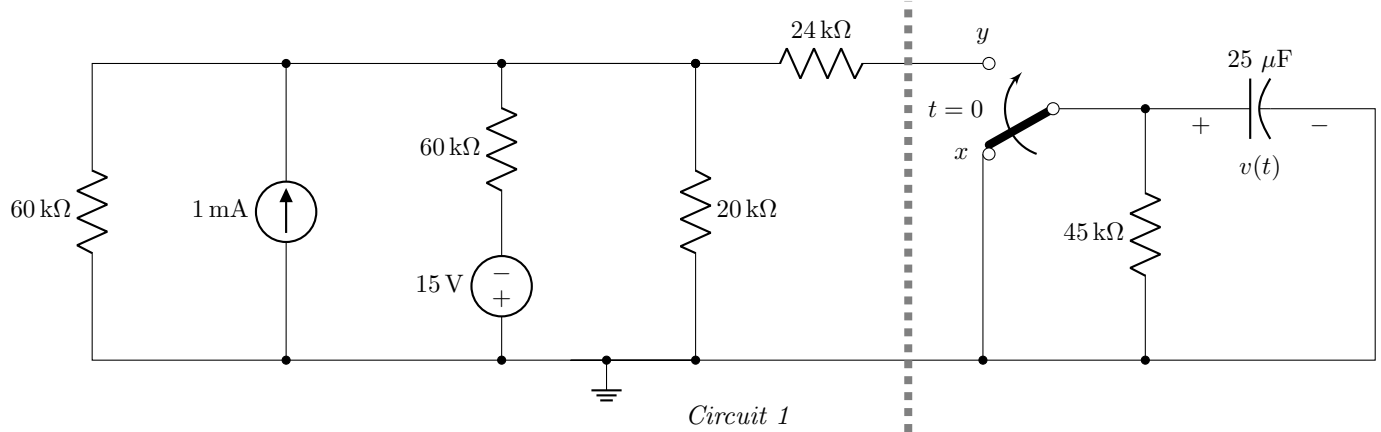
■ Question 2 of 3 [CO4 CO5] [20 marks]

- (a) The $v_c(t)$ vs t plot below shows the voltage response of a capacitor (C) in a series RC circuit to a sudden change in the DC voltage applied through an equivalent resistance of $4\text{ k}\Omega$.



- (i) [2 marks] **Determine** the approximate **Time Constant** from the figure. **Determine C** with appropriate unit.
- (ii) [1 mark] Write a mathematical expression of $v_c(t)$ for $t > 0$.
- (iii) [2 marks] Predict and draw a circuit with appropriate switching mechanism that can generate the voltage response as shown in the plot.

(b) Consider the following circuits.



- (i) [7 marks] **Reduce** the left portion with respect to the dashed gray line of *Circuit 1* so that it takes the form of *Circuit 2* as shown. Write down the values of V_1 and R_1 .
- (ii) [8 marks] Now, **analyze** the **Transient Behavior** of the circuit assuming that the switch moves from position x to position y at $t = 0$. **Determine $v(t)$** for $t > 0$.

■ Question 3 of 3 [CO4 CO6] [15 marks]

Determine $v_x(t)$ in the circuit shown below.

